

name: Hero Model Reset
overview: "Validate the mental reset: three kinds of model, locked hero as fixed target, and the simplest score→select story that can bend a selection curve. Vocabulary locked Jul 2026 — environment ≠ advancement; score ≠ select; assignment uses ρ (not τ)."
todos:
- id: memo-three-models
content: "Draft 1-page memo: phenomenological LPM vs L_net theory vs generative sim — what each proves and does not"
status: completed
- id: hero-estimand-lock
content: Confirm hero spec in writing (16 quantile, winsor 0.01–0.99, min_minutes=20, poolq_loo) as fixed target for all model tests
status: completed
- id: lpm-layer-a
content: Run quadratic LPM on hero-filtered panel; record coef signs and overlay vs ventile bins
status: completed
- id: pass-a-lambda-knockouts
content: "Pass A: λ>0 vs λ=0 scoring knockout; same 16-bin readout (hero_model_reset_bundle → alex_side_by_side_v0)"
status: completed
- id: alex-side-by-side
content: Produce side-by-side figure + nesting-note limitation sentence for Alex (empirical hero + generative POC)
status: completed
- id: pass-b-rho-ablation
content: "Pass B: ρ ablation (low/moderate/high + sort-and-chop) via 540_rho_ablation_bundle → alex_rho_ablation_v0"
status: completed
- id: defer-stretch
content: Explicitly park bin-for-bin LOO match and preferential attachment as stretch, not v1 gate
status: completed
isProject: false

Hero inverted-U: model types, testing, and smart restart

Synced vocabulary: 2026-07-28 — matches `BINDING_Selection_is_its_own_step.md` and `re_entry/02_Three_Kinds_of_Model.md`.
Living sim lab: `sports/540_READ_ME_SIM.md` / `540_three_step_sim.ipynb` (538D archived under `sports/archive/`).

Short answer: yes, your approach makes sense

You are doing the right thing by stepping back. The project distinguishes **phenomenon → theory → generative POC** (`05_Model_Nesting_Note_v1.md`, `30_Alex_Gates_inform_status_outline.md`). Confusion is mostly **which layer you are trying to “solve” at once**, plus older docs that said “selection” for the score equation. Splitting layers — and **score ≠ select** — is the smart move.
Re-entry path if lost: `re_entry/00_READ_ME_FIRST.md` → 01 → 02 → 03.

1. What “mathematical model” means here (three different objects)

In everyday conversation we say “the model” as if there is one thing. In this project there are **three different objects**, and mixing them up is the main source of confusion. Each answers a different question. None of them, by itself, is the whole dissertation story.

The three layers in plain language

Layer	Plain-language name	What you are actually doing	What a successful result looks like
A. Phenomenological / reduced form	"Describe the curve in the data"	Fit a simple functional form to real player-season rows — e.g. does draft probability bend downward at the top when you plot it against leave-one-out pool quality?	You can show, on the locked hero spec , that draft rate rises and then dips (or is concave) when you smooth or bin the empirical panel. You get coefficients and a picture.
B. Structural / mechanism (theory)	"Explain why help and hurt can coexist"	Write down a story in symbols: net learning has a benefit part and a congestion part; scoring may penalize crowding among viable peers, not just reward raw ability; then a winner rule picks who advances.	The story is internally consistent, maps to named objects in the code (A_i , congestion L_C , weight λ , winner rule top K), and suggests ingredient tests that can falsify the claim (e.g. "if congestion does not enter the score , the sim story fails").
C. Generative / simulation - Data Generating Process (DGP)	"Build a fake league and see if the rule produces a bent curve"	Simulate rosters: draw ability, assign players to teams (p), score candidates (S_i), select top K, then bin and plot who got selected vs a pool statistic.	Under a minimal rule set (score includes congestion, scarce slots), the shape of the simulated selection curve differs from talent-only scoring — e.g. elite bins compress relative to ability-only.

Do not use one word ("the model") for all three. Layer A is a **statistical summary** of MBB data. Layer B is **economic/statistical narrative** about mechanisms. Layer C is a **computerized thought experiment** that operationalizes part of Layer B.

Layer A — phenomenological (the hero and the quadratic LPM)

Question it answers: *Does an inverted-U-shaped (or concave) relationship between pool quality and draft probability show up in the real NCAA→NBA panel on a fixed, pre-specified definition of "quality"?*

What it is not: A claim that NBA teams literally maximize a particular formula, or that we have identified *why* the curve bends.

Concrete objects in the repo:

- Hero plot** — binned mean draft rate (**Y_draft**) vs **poolq_loo** : each player’s leave-one-out mean teammate performance (teammates’ box-score quality with that player removed so they are not counted twice).
- Quadratic LPM (linear probability model)** — OLS on the same filtered rows (not linear programming):

$$Y_{draft} \sim \beta_0 + \beta_1 \cdot poolq_loo + \beta_2 \cdot poolq_loo^2$$

wired in **panel_build.py** (**draft_poolq_quadratic_coeffs** , ventile overlay in 530).

Why keep the quadratic: $\beta_2 < 0 \Rightarrow$ fitted curve **concave down**; one turning point $Q^* = -\beta_1/(2\beta_2)$ (local max if $\beta_2 < 0$). No inflection on a pure quadratic. Claims are about the **fit**; binned hero stays the stylized fact. See also **re_entry/02** Layer A.

What "test" means here: Check sign and magnitude of β_2 (negative $\Rightarrow$ concave / inverted-U tendency); compare the smooth curve to **ventile bin means** visually; optional out-of-sample checks. You are **fitting the phenomenon**, not simulating a league.

What Layer A proves: The stylized fact is **real on the locked estimand** (16 quantile bins, winsorized **poolq_loo** , **min_minutes=20** , etc.).

What it does not prove: Mechanism, causal effects, or that any NBA front office uses this score.

Layer B — structural / mechanism (theory spine)

Question it answers: *Why is it plausible that being on a "good" team pool could both help and hurt draft odds?* — and **why describing the environment is not the same as deciding who advances**.

What it is: Prose and notation — not one estimated equation in v1. **Keep three separations** (BINDING):

Part 1 — Environment (**L_net = B(·) - D(·)**): Peers confer benefit (B) and congestion (D) on net development / value in the environment. This **describes the pool**, not who gets drafted.

Part 2 — Scoring (Alex score): **S_i = A_i - λ · L_C** . Congestion enters **ranking**, jointly with ability **A_i** . **λ lives here**. This is **not** the full reduced-form environment and **not** the hero regression.

Unified nest (Alex v1): With **(B - D) = -L_C** in the congestion channel used for ranking,

$$S_i = A_i + \lambda(B - D) = A_i - \lambda \cdot L_C.$$

Knockout: $\lambda = 0 \Rightarrow$ **S_i = A_i** (talent-only score).

Part 3 — Selection (winner rule): Given scores, **who wins?** v1 default = **top K**. Later can be soft / noisy / stochastic. **Do not call** **S_i** "selection." Score $\neq$ select.

Charles binding (Jul 2026): **BINDING_Selection_is_its_own_step.md**

What "test" means here: Narrative consistency + sim **scoring knockouts** — e.g. $\lambda = 0$ (ability-only score, same top-K) and ask whether the generative story still holds.

What Layer B proves: A coherent **mechanism vocabulary** and falsifiable **directional** claims (congestion in the **score** can matter).

What it does not prove: Separate identification of **B(Q)** and **D(Q)** as functions of one axis in v1, or that the real NBA implements **S_i** literally.

Layer C — generative simulation (the DGP)

Question it answers: *If we write down an explicit **data-generating process** — artificial players, teams, and a draft rule — does a **minimal** congestion term in the **score** change the **shape** of who gets picked when we bin by pool quality?*

DGP = data-generating process: the full recipe that **creates** synthetic rosters and draft outcomes, as opposed to fitting a curve to real outcomes after the fact.

Three-step pipeline (locked):

- 1. **Assign** — soft match ability to team types T_j with user-facing assortativity p (σ fixed ~ 0.65). Preferential attachment = optional boolean, default **off**. Legacy docs used τ for temperature (opposite intuition) — treat τ as **archived notation only**.
- 2. **Score** — e.g. $S_i = A_i - \lambda \cdot L_C$ (or ability-only when $\lambda = 0$). Soft crowding inside L_C (viability sigmoid) is **not** assignment softness and **not** selection noise.
- 3. **Select** — **top K** draft slots (scarce winners). Later: stochastic draw from scores without renaming the score “selection.”

Concrete stack in the repo:

- **Living lab:** `540_READ_ME_SIM.md`, `540_three_step_sim.ipynb`, engines under `sports/sports_pipeline/tier1_*.py`.
- **Pass A exports:** `hero_model_reset_bundle.py` → `alex_side_by_side_v0/` (λ knockout).
- **Pass B exports:** `540_rho_ablation_bundle.py` → `alex_rho_ablation_v0/` (p levels + sort-and-chop diagnostic).
- **Archived:** 538D and sibling labs under `sports/archive/` — historical source of Pass A lessons; not the open workspace.

What “test” means here: Re-run the DGP with knobs toggled; compare **curves**, not regression R^2 on real data. Headline knockout (Pass A): $\lambda > 0$ (congestion in score) vs $\lambda = 0$ (talent-only score), **same top-K** — does the elite tail behave differently?

What **Layer C** proves: A **minimal artificial league** where talent-only **scoring** is **not enough** and a congestion term in the score **can** bend who gets selected — proof-of-concept for Alex’s mechanism.

What it **does not prove**: Bin-for-bin replication of every hero ventile rate, or that the real draft is this sim.

How the three layers relate (read this before the diagram)



Reading the diagram:

- **Top path (Rung 1):** Start from **real MBB data** → hero bins + optional quadratic overlay. This is Layer **A**. No simulation required.
- **Middle path (Rung 2):** Start from **rules** → assign → score → select → binned plot. This is Layer **C**. The sim **tests the rule**, not the regression.
- **Theory box:** Layer **B** sits **between** interpretation and design — it explains why **scoring** might use $A - \lambda L_C$, not only A , but it does not by itself produce numbers until wired into Layer C (or decomposed empirically later at Rung 2.5).

Alex side-by-side (what v1 actually delivers): Layer **A** hero PNG + Layer **C** sim PNG, same **Y** (advancement / draft rate), **honest X** labels (empirical LOO quality vs sim bins on the best available matching axis), plus an explicit **limitation sentence** — not “derive the hero from a closed-form integral.”

One paragraph to keep in memory

Layer **A** can reproduce an inverted-U **without telling you why** — it is curve-fitting and binning on the real panel (**outcomes only**). Layer **B** splits **environment** ($B - D$) from **advancement** (score then select). Layer **C** runs **assign** → **score** → **select** in code and shows talent-only **scoring** is insufficient under a fixed winner rule. You need **A** for the empirical fact, **B** for the dissertation logic (environment $\neq$ advancement; score $\neq$ select), and **C** for the claim that **congestion in the score** can bend curves — three jobs, three objects.

2. Does simulation test a formula?

Yes — when the formula is a **generative rule**, not when it is only a regression fit.

- **Simulation tests:** “If we **score** with $S_i = A_i - \lambda \cdot L_C$, **select** top K, and rosters form this way, does **P(selected)** bend non-monotonically when we bin by a pool statistic?”
- **Regression tests:** “Does $P(\text{draft} \mid \text{poolq_loo})$ look quadratic in the **real panel**?” (already wired in 530)

Both are legitimate; they answer **different questions**. Simulation is **not** required to *draw* an inverted-U (a quadratic fit can do that). Simulation is required for the Wang-style claim: **talent-only scoring is not enough; a congestion term in the score changes who gets selected** (Pass A: $\lambda=0$ vs $\lambda>0$).

3. Your two simplification strategies — which to use when

(a) Top-down: full mechanism → drop until it breaks

Best for: Layer B/C — Alex score, assignment, winner rule.

Procedure: 1. Start with the **full generative stack** you already built (ability draw, soft assign with ρ , congestion in score, top-K). 2. Fix **one empirical anchor** (hero spec below). 3. Remove or zero one ingredient at a time and record **what breaks**: - $\lambda = 0$ → talent-only **score** (should flatten / fail congestion story) — **Pass A** - change ρ (or sort-and-chop diagnostic) → changes pool overlap / tail bins — **Pass B** - no L_C term, only A_i → scoring on ability alone

This is **mechanism identification in the sim**, not coefficient estimation on real draft data.

(b) Bottom-up: minimal curve → decompose

Best for: Layer A — describing the hero.

Procedure: 1. Start with **simplest curve**: $E[Y \mid Q] = \beta_0 + \beta_1 Q + \beta_2 Q^2$ with $\beta_2 < 0$ (530 LPM). 2. Decompose Q ($poolq_loo$) into interpretable pieces only when moving to Rung 2.5: - $poolq_loo \approx$ quality column (B-ish mix) - $crowding_smooth \approx$ congestion column (D-ish) — CELL 5d in [530_sports_pipeline.ipynb](#)

Recommendation: Use **both**, in order: **bottom-up for the hero (A)**, **top-down for the sim (C)**. They are not competing; they are different rungs.

4. Assortative grouping, ρ , etc. — simulation details or essentials?

Mostly simulation / DGP details — not the core mathematical claim.

Knob	Role	Needed for v1 Alex side-by-side?
λ / congestion in score	Core mechanism ingredient	Yes (Pass A done)
Top-K selection	Scarcity of draft slots (winner rule)	Yes (conceptually)
Soft assignment (ρ , T_j)	How players land on teams	Helpful for realistic pool overlap; not the headline theorem (Pass B diagnostic done)
Sort-and-chop benchmark	Hard assortative diagnostic	Diagnostic only — not $\rho \rightarrow \infty$
Preferential attachment	Boolean roster feedback	Off by default; stretch
Empirical team distribution on/off	Faithful historical sweeps	Stretch / amplitude — not gate for “minimal POC”

Notation trap (archived): Older plans/labs used τ for assignment temperature. **User-facing knob is ρ (assortativity)**. Soft crowding parameters inside L_C are a different softness — do not reuse “ τ ” for both.

Path II (locked in nesting note): generative figure may use **pool mean** on X while empirical hero uses $poolq_loo$. Say that aloud; do not over-fit assortativity until the **minimal** scoring story works.

Stretch goal (parallel, not v1 gate): match hero on $poolq_loo$ with same binning ($alex_side_by_side_v0/$ spec: 16 quantile, winsor 0.01–0.99, $min_minutes=20$).

5. Do you need simulation? Why?

Goal	Need sim?
Show inverted-U exists in MBB data	No — hero is done
Fit a smooth curve through bins	No — quadratic LPM
Explain <i>why</i> help and hurt can coexist	Partly prose ($L_{net} = B - D$)
Show congestion in the score is a plausible minimal ingredient	Yes — sim with λ knockout (Pass A / 540 stack): congestion in score vs talent-only, same top-K
Prove identification of B(Q) and D(Q) separately on one axis	No — explicitly deferred in v1

Why sim exists in your project: Alex’s score is a **ranking rule for an artificial league**, not a claim that real NBA teams optimize S_i . Simulation is how you **operationalize and stress-test** that rule (then apply a winner rule) — including the **knockout** that can falsify the congestion-in-score claim. It is the supplement; the hero is the spine.

6. What is probably a “keeper” vs what can wait

Keep (solid foundation): - Locked hero triplet + 530 panel/LOO pipeline - Nesting note + Alex brief axis honesty + BINDING + re_entry 00–03 - **540** three-step sim + Pass A ($alex_side_by_side_v0/$) + Pass B ($alex_rho_ablation_v0/$) - Sensitivity lessons in

[Pertinent_Thoughts_Scout.md](#) ([min_minutes](#) , quantile vs equal-width) - Archived 538D lessons (do not reopen as daily workspace)

Defer / shrink (avoid re-blocking): - Bin-for-bin LOO replication as **v1 requirement** - Large faithful historical sweeps on Rivanna until minimal score→select story stays locked - HS binning, multiplicative rewrite, full [B\(Q\)–D\(Q\)](#) estimation - Preferential attachment on; denser assortativity grids beyond Pass B diagnostics

7. Recommended sequence (status as of Jul 2026 sync)

1. **Write one page “three models” memo** — **done** ([Hero_Model_Three_Layers_Memo.md](#) ; re_entry 02 is the Charles-facing version).
 2. **Freeze hero estimand** — **done**: 16 quantile, ppm z-scored, winsor (0.01, 0.99), [min_minutes=20](#) , 2011–2021, [poolq_loo](#) , draftee filter off.
 3. **Layer A check** — **done**: quadratic LPM on same filtered rows.
 4. **Pass A (λ knockout)** — **done**: [\$\lambda>0\$](#) vs [\$\lambda=0\$](#) , same bin count; side-by-side in [alex_side_by_side_v0/](#) .
 5. **Pass B (ρ ablation)** — **done**: low / moderate / high [\$\rho\$](#) + sort-and-chop diagnostic in [alex_rho_ablation_v0/](#) .
 6. **Alex slide**: side-by-side PNGs + one limitation sentence from nesting note §4 — **ready**.
 7. **Only then** revisit preferential attachment / faithful sweeps if LOO match still matters scientifically.
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8. One-sentence answers to your numbered questions

1. **Recreate inverted-U**: Yes — via **quadratic reduced form** (easy) and/or **generative scoring with congestion + top-K select** (mechanism); full [B–D](#) decomposition is theory, not yet a single estimated formula.
2. **What is a model / how tested**: Three layers (table above); simulation tests **generative rules** (assign → score → select), regression tests **empirical curve**, theory frames **interpretation**.
3. **Top-down vs bottom-up**: Use **both** — bottom-up for hero fit, top-down for sim mechanism stripping.
4. **Assortativity ([\$\rho\$](#)) etc.**: Mostly **simulation plumbing**; not the v1 theorem. Sim is needed for **minimal mechanism POC**, not for proving the hero exists.
5. **Reset smart? Yes**. Keep infrastructure; re-order claims around rungs; stop requiring one formula to do all jobs; keep **score $\neq$ select** in every write-up.
6. **Approach correct? Yes**, with one guardrail: **do not make bin-for-bin LOO sim match the gate for “we have a model.”** That is north-star R&D, not the definition of success for the next Alex conversation.